

1. What Went Well

During Sprint 3, the team successfully improved both the functionality and usability of the system.

We were able to implement key features such as authentication, role-based dashboards, UI/UX improvements, and explainable AI integration. The overall system became more structured and closer to a real-world clinical application.

Team collaboration was effective, and tasks were clearly divided among members. The use of GitHub for issue tracking and branch management helped in organizing development work and maintaining progress.

We also successfully completed frontend modularization, which improved code maintainability and made further development easier.

Additionally, the integration of explainability features (SHAP-based insights) was a major achievement, as it enhanced the transparency of the model predictions.

2. What Could Be Improved

Although the sprint was successful, there are some areas where improvements can be made.

Some UI inconsistencies appeared during development due to multiple changes happening in parallel. These could be reduced by defining a design standard earlier.

Merge conflicts occurred while integrating multiple feature branches, especially in frontend files. This indicates a need for better coordination and more frequent merging.

Time management can also be improved, as some tasks such as final UI cleanup and testing were completed towards the end of the sprint.

Additionally, better planning for integrating explainability features could have reduced the complexity faced during implementation.

3. What We Learned

During this sprint, we gained valuable experience in both technical and team collaboration aspects.

We learned how to design and implement a modular frontend structure, which is important for building scalable applications.

We also understood how to integrate explainable AI into a real system and present model outputs in a meaningful way for users.

From a development perspective, we improved our understanding of Git branching strategies, pull requests, and resolving merge conflicts.

We also learned the importance of user-centered design while developing healthcare-related applications, ensuring clarity, usability, and reliability.

4. Action Items for Next Sprint

Based on the Sprint 3 experience, the following improvements will be carried forward to Sprint 4:

- Define UI design standards before implementation
- Perform more frequent branch merges to avoid conflicts
- Allocate dedicated time for testing and validation
- Deploying the project in cloud platform.

5. Team Contribution Summary

During Sprint 3, tasks were clearly distributed between team members based on frontend, backend, and integration responsibilities.

- **Ravi**
Worked primarily on core system enhancements and explainability features. This included implementing SHAP-based model explainability, enhancing the prediction API, improving input validation and error handling, designing the prediction result page with explanation outputs, and handling prediction logging. Ravi also contributed to UI/UX improvements for the assessment form and was responsible for Sprint 3 planning, project board updates, and documentation.
- **Uday**
Focused on authentication and system integration tasks. This included enhancing the login system with standard features, implementing forgot password functionality, and connecting the frontend with the backend prediction API. Uday also played a key role in end-to-end system integration testing to ensure all components worked together correctly. Additionally, Uday collaborated on improving the assessment form UI/UX.

The team collaborated during integration and testing phases to ensure smooth functionality and successful completion of all Sprint 3 tasks.